
Federated Learning for Privacy-Preserving Distributed Space Weather Monitoring in Smart Grid Protection

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Abstract — The vulnerability of modern power grids to geomagnetically induced currents (GICs), triggered by M- and X-class solar flares, presents a critical and recurring threat to bulk electric infrastructure; the 1989 Québec system collapse remains the canonical demonstration of how a single geomagnetic storm can disable an entire provincial grid. Effective early-warning systems require predictive models trained on high-fidelity solar observations spanning multiple geographic regions, yet nationally sensitive space-weather data collected by agencies such as NASA, ESA, JAXA, ISRO, KASI, and the Australian Bureau of Meteorology cannot be centralised because of data-sovereignty constraints and inter-agency data-sharing agreements. This paper presents, to the best of our knowledge, the first application of federated learning (FL) to solar flare prediction for smart-grid protection, explicitly addressing this privacy barrier. The proposed framework partitions the SWAN-SF benchmark dataset into six geographically distributed regional-observatory clients using Dirichlet sampling, inducing realistic non-IID label skew. Each client trains a local neural model on 144 statistical features derived from 24 photospheric magnetic-field parameters, and transmits only weight updates—never raw satellite measurements—to an aggregation server that combines FedAvg and FedProx (proximal coefficient $\mu = 0.01$) with distribution-aware weighting. On a held-out test set that preserves the natural $\sim 1.9\%$ flare prevalence, FedProx attains an ROC-AUC of 0.930 (95.0% of the centralised XGBoost upper bound of 0.978), a recall of 0.416, and an F1-score of 0.326 at an F_β -optimal decision threshold ($\beta = 2$), whereas FedAvg degenerates under the same partition. SHAP analysis of the centralised surrogate identifies extrema of the flux-emergence proxy (R-value), current helicity, and Lorentz-force variability as the dominant predictors, consistent with established solar physics. The framework converges within 50 communication rounds, making it operationally compatible with existing SCADA-based grid-management infrastructure. These results demonstrate that distributed space-weather agencies can jointly train a usable flare-prediction model without compromising data sovereignty, establishing federated machine learning as a practical mechanism for global smart-grid resilience against geomagnetic threats.

Keywords: federated learning; solar flare prediction; smart grid; geomagnetically induced currents; FedProx; space weather; SWAN-SF; non-IID data; class imbalance; data privacy

Source code and all experimental artefacts: <https://github.com/Daveshvats/Federated-Learning-for-Distributed-Space-Weather-Monitoring>

1 Introduction

Modern society has quietly built its electrical backbone in the path of a cosmic hazard. Geomagnetically induced currents (GICs) are quasi-direct currents that flow in high-voltage transmission

lines when a geomagnetic storm—itsself driven by the arrival of a coronal mass ejection (CME) associated with a major solar flare—distorts the Earth’s magnetosphere. The time-varying geomagnetic field induces geo-electric fields in the conducting crust, and these fields drive GICs through grounded transformer neutrals. Once inside a high-voltage transformer, even a modest GIC biases the alternating flux into half-cycle saturation, producing harmonics, reactive-power absorption, and localized heating that can culminate in irreversible thermal damage [1, 2]. The canonical demonstration of this failure mode remains the 13 March 1989 storm, which collapsed the Hydro-Québec grid in ninety seconds and left six million customers without power for nine hours [1, 2]. The July 2012 event, an eruptive flare/CME of Carrington-class magnitude that missed Earth’s orbital position by roughly nine days, is a stark reminder that the next exposure is a matter of when, not if [3].

The first line of defence against such events is forecasting. Grid operators who receive an early warning that an M- or X-class flare has erupted, or is likely to erupt, in a magnetically complex active region can pre-emptively derate transformers, adjust reserve margins, postpone switching operations, and prepare load-shedding plans in the window between the flare observation and the CME’s arrival one to three days later. Machine-learning flare-prediction models built on photospheric vector-magnetogram summaries—the SHARP parameter suite [4] and its descendants—have demonstrated statistically skilful classification of flaring versus non-flaring active regions [5, 6]. The predictive signal lives in quantities such as total unsigned current helicity, magnetic free energy, and flux-emergence proxies, all of which are computed routinely by solar observatories on every continent and in orbit.

There is, however, a structural obstacle to training the strongest possible model: the data cannot be pooled. The highest-fidelity flare-forecasting observations are produced and curated by nationally mandated agencies—NASA and NOAA in the Americas, ESA in Europe, JAXA in the Asia-Pacific region, ISRO in South Asia, KASI in East Asia, and the Australian Bureau of Meteorology (BoM) in Oceania. Raw magnetogram-derived time series are treated as nationally sensitive assets in many jurisdictions, and inter-agency data-sharing agreements historically favour metadata exchange over bulk data transfer. Consequently, each agency trains (or could train) models on its own regional shard, and the global community inherits the fragmentation of six models each seeing a biased slice of the active-region population. A genuinely global model, trained on all regions’ data without any region surrendering that data, has been out of reach.

Federated learning (FL) [7, 8] is precisely the mechanism for this setting. In cross-silo FL, each data-holder trains a shared model architecture locally and transmits only parameter updates to an aggregation server, which averages them into a new global model. Raw measurements never leave the custodian’s infrastructure. FL has already proven transformative in medicine and digital health, where patient-privacy constraints are structurally identical to the data-sovereignty constraints of space-weather agencies [9]. What has not existed, prior to the present work, is a systematic treatment of FL for solar flare prediction—the solar-physics community’s benchmark workflows [5, 6, 10] are all centrally pooled, and the FL literature contains, to the best of our knowledge, no application to photospheric flare forecasting.

This paper closes that gap. We present a complete, reproducible federated framework—implemented, versioned, and released as open source—that simulates six geographically distributed regional observatories, each holding a non-IID shard of the SWAN-SF benchmark [10] in its cleaned and rebalanced form, and evaluates whether federation can approach the centralised upper bound without any data movement. The specific contributions are:

1. **A problem formulation and system architecture** that maps smart-grid GIC early warning onto a cross-silo federated classification problem with a safety-critical, recall-weighted objective, and that abstracts the data-sovereignty constraint of national space-weather agencies into a concrete, simulable protocol (Section 3 and Section 4).

2. **A non-IID federation protocol for SWAN-SF** that combines Dirichlet label-skew partitioning across six regional clients, per-client SMOTE oversampling, a server-aware Fed-Focal loss [11] for extreme class imbalance, and distribution-aware aggregation weighting, on top of both FedAvg [7] and FedProx [12] training loops (Section 4).
3. **A safety-critical evaluation protocol** based on F_β threshold optimisation with $\beta = 2$, which recognises that in grid protection a missed flare is operationally far costlier than a false alarm, and an end-to-end comparison against centralised logistic-regression and XGBoost baselines under the natural test-set prevalence of approximately 1.9% flares (Section 5).
4. **An empirical privacy-cost quantification.** On the held-out test partitions, FedProx achieves an ROC-AUC of 0.930 against the centralised XGBoost upper bound of 0.978 (95.0% of pooled discrimination), with recall 0.416 and F1 0.326 at the recall-weighted threshold, while FedAvg collapses under the same partition—evidence that proximal regularisation, not merely federation, is what makes the non-IID setting tractable (Section 6).
5. **An interpretability audit.** SHAP analysis of the centralised surrogate, with a decoding of the statistical feature space back to physical magnetic parameters, identifies extrema of the Schrijver flux-emergence proxy [13], current helicity, and Lorentz-force components as the dominant predictors, consistent with established flare physics (Section 6).

In addition to these scientific contributions, the released framework embodies a documented engineering narrative: a six-version stability iteration (v2.1–v2.6) in which Dirichlet concentration, focal-loss alpha clamping, optimiser choice for variance-reduced FL, and dataset auditing were each adjusted in response to diagnosed failure modes. We summarise this evolution in Appendix A because it constitutes, in our view, transferable know-how for practitioners attempting the first FL deployment in any physical-science domain.

The remainder of the paper is organised as follows. Section 2 reviews flare prediction and federated learning. Section 3 formalises the problem. Section 4 details the system architecture and methodology. Section 5 describes the experimental setup, Section 6 presents and analyses results, and Section 7 discusses operational integration and engineering lessons. Section 8 states limitations and future work, and Section 9 concludes.

2 Background and Related Work

2.1 Solar flare prediction from photospheric magnetograms

The physical premise of data-driven flare forecasting is that the photospheric magnetic field of an active region encodes the stress budget that a future flare will release. Schrijver [13] demonstrated that a single scalar—the R-value, the unsigned flux within high-gradient polarity-separation corridors—separates major-flaring regions from non-flaring ones with striking reliability, and this proxy remains among the strongest single flare predictors identified to date. Bobra and Couvidat [4] operationalised the machine-learning view of the problem, showing that a support-vector machine trained on 25 SHARP parameters derived from SDO/HMI vector magnetograms can forecast M-class flares with useful skill, and that feature selection matters as much as model choice. Subsequent community-wide blind tests, most notably the multi-method comparison of Leka et al. [5], established two durable findings: (i) no single method dominates across all event definitions and skill metrics, and (ii) the effective ceiling on skill is set by the quality and homogeneity of the training data rather than by classifier complexity.

The European FLARECAST project [6] pushed the operational agenda further, combining multiple forecasting methods and data sources in an open framework, and it explicitly confronted

the data-heterogeneity problem that motivates the present paper. On the data side, the SWAN-SF benchmark of Angryk et al. [10] released a multivariate time-series dataset of roughly 8,000 active-region tracks with 24 physical parameters per timestep and flare-history labels, specifically designed to make flare prediction reproducible and comparable; it has since become the community’s reference benchmark, and a family of cleaned and rebalanced variants (random undersampling, Tomek-link cleaning, and TimeGAN-based synthetic augmentation [14]) has been curated to make the benchmark tractable for supervised learning. Recent deep-learning work on SWAN-SF continues to raise the centralised ceiling: contrastive pretraining and multifaceted preprocessing [15] and LSTM-style temporal architectures [16] exploit the sequential structure of the 60-minute observation windows. All of these works, without exception, assume that the full training corpus is available in one place.

2.2 Federated learning under statistical heterogeneity

Federated learning was introduced by McMahan et al. [7] in the FedAvg formulation: clients perform several epochs of local SGD on private data, and a server aggregates the resulting weights by sample-size-weighted averaging. The defining difficulty, identified almost immediately, is statistical heterogeneity: when client data distributions differ—the non-IID setting—local optima diverge from the global optimum, and naive averaging can oscillate or diverge [17]. FedProx [12] addresses this by adding a proximal term $(\mu/2) \|\mathbf{w} - \mathbf{w}^{(t)}\|^2$ to each client’s local objective that anchors local training to the current global model, tolerating partial work and heterogeneous local computation. SCAFFOLD [18] attacks the same problem with control variates that correct client drift using variance reduction. The comprehensive survey of Kairouz et al. [8] situates these algorithms within the broader open-problem landscape, including robust aggregation, personalisation, and privacy. Cross-silo FL—few, reliable, institutionally owned clients—is the deployment regime relevant to space-weather agencies, and it contrasts with the cross-device regime in its stability requirements and long-lived participants.

2.3 Class imbalance in federated learning

Flare prediction inherits an extreme class-imbalance regime: in SWAN-SF’s natural test distribution, roughly 1.9% of windows precede M/X-class flares, while the rebalanced training partitions sit near parity. The focal loss of Lin et al. [19], originally developed for dense object detection, down-weights easy examples with a $(1 - p_t)^\gamma$ modulating factor and concentrates gradient on hard positives; it has become the default imbalance workhorse. Sarkar et al. [11] adapted focal loss specifically to FL in the Fed-Focal formulation, in which the class-weighting parameter is adjusted using information available at the server about the global class distribution, so that clients with different local positivity rates do not implicitly reweight the global objective. In parallel, synthetic oversampling remains a standard pre-processing response: SMOTE [20] interpolates minority-class neighbours, and mixup [21] creates convex combinations of training pairs. Under federation, rebalancing must be applied locally per client, since the pooled distribution is precisely what no participant can inspect.

2.4 Positioning of this work

Two research threads—physics-grounded flare prediction and heterogeneity-robust federated learning—have developed largely independently, and we are not aware of any prior work that joins them. Searches of the FL literature reveal applications spanning digital health [9], finance, and IoT sensing, but none addressing space-weather forecasting or solar flare prediction; conversely, the flare-forecasting literature, including its most recent deep-learning incarnations [15, 16], is entirely centralised. The present work is, to the best of our knowledge, the first

federated treatment of solar flare prediction, the first FL framework evaluated on the SWAN-SF benchmark, and the first to quantify the privacy cost of federation—the gap between federated and centralised skill—specifically for a safety-critical smart-grid application. In doing so it also contributes a rarely documented artifact: an honest engineering log of the failure modes encountered when FL meets a severely imbalanced physical-science benchmark (Appendix A).

3 Problem Formulation

3.1 Federated risk minimisation

Let K denote the number of participating agencies (clients), each holding a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of active-region observation windows $x_i \in \mathbb{R}^d$ and binary labels $y_i \in \{0, 1\}$ indicating whether an M- or X-class flare occurs within the forecasting horizon. Let $\mathcal{D} = \bigcup_{k=1}^K \mathcal{D}_k$ with $n = \sum_k n_k$. The goal of standard centralised learning is to solve

$$\min_{\mathbf{w} \in \mathbb{R}^p} F(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\mathbf{w}}(x_i), y_i), \quad (1)$$

where $f_{\mathbf{w}}$ is the prediction model, ℓ a per-sample loss, and p the number of parameters. In the federated setting this objective is decomposed as a weighted sum of local empirical risks

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^K \rho_k F_k(\mathbf{w}), \quad F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i), \quad (2)$$

with aggregation weights $\rho_k \geq 0$, $\sum_k \rho_k = 1$. The vanilla FedAvg choice is $\rho_k = n_k/n$; Section 4 describes the distribution-aware weights used in this work.

3.2 The privacy constraint

The operational constraint that distinguishes cross-silo FL from distributed optimisation is not an algorithmic preference but an institutional invariant: for every round t and client k , the server may observe only the weight increment $\Delta \mathbf{w}_k^{(t)} = \mathbf{w}_k^{(t+1)} - \mathbf{w}_k^{(t)}$ (or equivalently the locally trained weights), never x_i or y_i for any sample in $\mathcal{D}_k$. Formally, the communication transcript

$$\mathcal{T} = \{\Delta \mathbf{w}_k^{(t)}\}_{k=1, \dots, K; t=1, \dots, T} \quad (3)$$

is the only quantity that crosses the agency boundary. This matches the data-sovereignty position of national space-weather agencies, for which raw magnetogram time series are nationally held assets, and it is the property that the released framework enforces by construction: the client-side pipeline never serialises raw shards.

3.3 Statistical heterogeneity

Because different agencies monitor different populations of active regions—different longitudes, hemispheres, and instrument footprints—the local class priors $p_k = \Pr(y = 1 \mid k)$ differ from the pooled prior $\bar{p}$. We simulate this label skew by partitioning the pooled training data among K clients with a symmetric Dirichlet distribution of concentration α (Section 4), so that the realised local priors span a wide range around $\bar{p}$. The consequence for optimisation is well documented [12, 17]: local objectives F_k become misaligned, and plain averaging of locally optimised weights introduces drift that grows with the number of local epochs. FedProx counteracts this drift by locally solving

$$\min_{\mathbf{w}} F_k(\mathbf{w}) + \frac{\mu}{2} \left\| \mathbf{w} - \mathbf{w}^{(t)} \right\|^2, \quad (4)$$

which keeps client updates in a trust region around the global model. The strength of this anchor, μ , is a central hyperparameter of this study.

3.4 A safety-critical, recall-weighted objective

Grid protection inverts the usual classification cost structure. A false-positive flare alert costs a operator a conservative derating or an unnecessary operational posture; a false negative can cost a fleet of high-voltage transformers. Following operational forecasting practice, we therefore evaluate with the F_β measure

$$F_\beta = \frac{(1 + \beta^2) P R}{\beta^2 P + R}, \quad \beta = 2, \quad (5)$$

where P and R are precision and recall; $\beta = 2$ weights recall twice as heavily as precision. Because the deployed decision rule is a threshold τ on the predicted flare probability, the framework treats τ as a free, per-model parameter selected on held-out validation evidence to maximise Eq. (5), rather than fixing $\tau = 0.5$ by convention. Section 6 shows this choice is not cosmetic: at the conventional threshold, every federated model trained on rebalanced local data is operationally unusable, and threshold optimisation is what converts a well-ranked model into a usable detector.

4 System Architecture and Methodology

4.1 Architecture overview

Figure 1 shows the complete system. Six regional observatory clients—stand-ins for NASA/NOAA, ESA (PROBA-2), JAXA, ISRO, KASI, and BoM—each hold a private, non-IID shard of the benchmark. In every communication round, each selected client trains a local copy of the shared model on its shard (with local rebalancing and an imbalance-robust loss, described below), and returns only its weight increment to the aggregation server. The server forms the next global model with distribution-aware weighted averaging, broadcasts it to all clients, and the cycle repeats for $T = 50$ rounds. Downstream of federation, the global model is calibrated for operations: its flare probabilities pass through the F_β -optimal threshold and become GIC early-warning decisions, which plug into existing SCADA workflows for grid management. The division of labour is deliberate: everything that touches raw magnetogram data (loading, rebalancing, local training) executes inside the client boundary; everything the server touches is a weight tensor.

4.2 Data: the SWAN-SF benchmark and its cleaned variant

All experiments use the SWAN-SF benchmark [10] (doi:10.7910/DVN/EBCFKM), a multivariate time-series dataset of active-region observation windows drawn from SDO/HMI vector magnetograms, with 24 physically motivated parameters per timestep sampled over 60-minute sliding windows and labels indicating the occurrence of $\geq M$ -class flares within the prediction horizon. The 24 parameters summarise magnetic-field complexity and include, among others: total unsigned current helicity (TOTUSJH), total photospheric magnetic free energy (TOTPOT), total unsigned vertical current (TOTUSJZ), total unsigned flux (USFLUX), the sum of absolute net-current-per-polarity (SAVNCP), absolute net current helicity (ABSNJZH), mean shear angle and the fraction of high-shear area (MEANSHR, SHRGT45), the Schrijver flux-emergence proxy (R-value), and the three components of the total Lorentz force (TOTFX, TOTFY, TOTFZ) [4, 13].

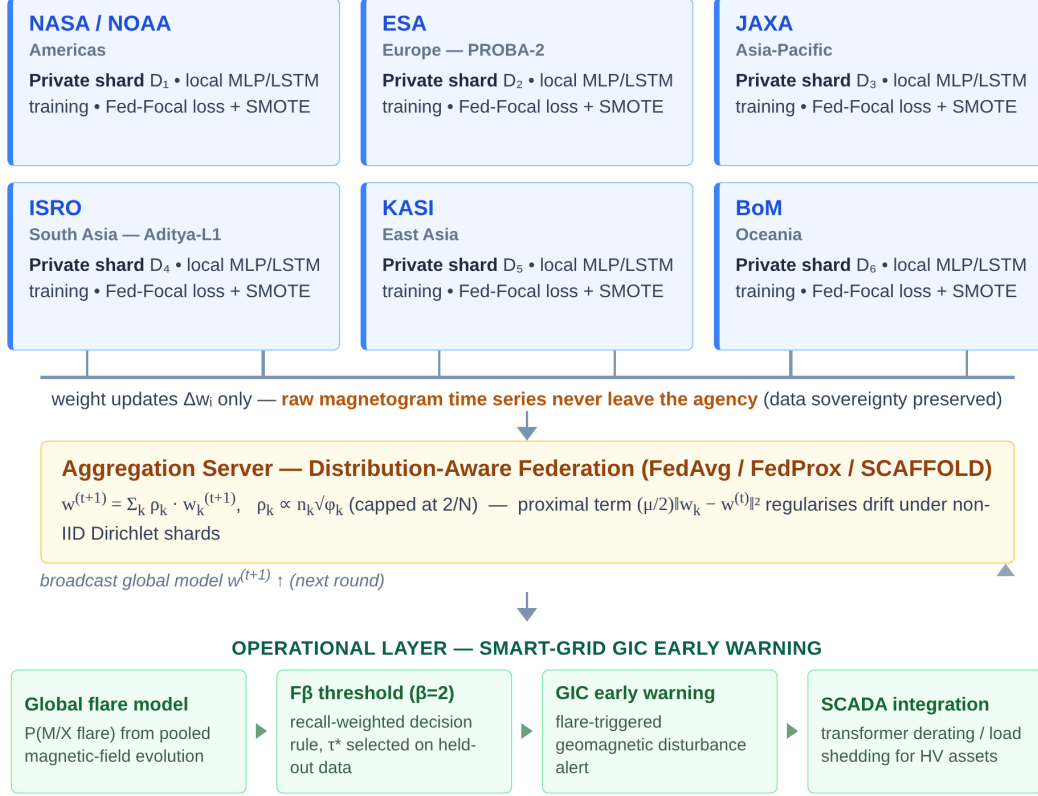


Figure 1: Federated space-weather monitoring architecture for smart-grid protection. Six regional observatory clients hold private non-IID shards of the benchmark; only weight updates $\Delta \mathbf{w}_k$ cross the agency boundary. The aggregation server applies FedAvg/FedProx with distribution-aware weighting ($\rho_k \propto n_k \sqrt{\phi_k}$, capped at $2/K$). The global model feeds an F_β -optimal decision layer for GIC early warning and SCADA integration.

We ingest the benchmark through its community-maintained cleaned variant¹, combining all five data partitions. The training partitions are class-balanced by this preprocessing (pooled positive rate $\bar{p} \approx 0.489$), while the untouched test partitions retain the natural, severely imbalanced prevalence of approximately 1.9% flare windows across roughly 3.3×10^5 held-out samples. This train–test contrast is not an artefact we introduce; it is inherited from the benchmark’s design philosophy, and it has a consequence that shapes the entire evaluation: models trained at $\sim 49\%$ prior must detect events at a $\sim 2\%$ prior, so absolute probability calibration degrades and decision thresholds must be re-optimised post hoc (Section 4.8).

4.3 Client-side feature representation

The framework supports two representations of the same 60-timestep observation windows. In the MLP path used for the reported run, each three-dimensional window $X \in \mathbb{R}^{60 \times 24}$ is reduced to a 144-dimensional statistical vector

$$\tilde{x} = [\mu, \sigma, \max, \min, \delta, s] \in \mathbb{R}^{144}, \quad (6)$$

where each block is applied per base feature: μ and σ are the temporal mean and standard deviation, max and min the window extrema, $\delta = X_T - X_1$ the net trend, and s the least-

¹Cleaned SWAN-SF Dataset, <https://github.com/samresume/Cleaned-SWANSF-Dataset>, which applies per-partition preprocessing: random undersampling with Tomek links and TimeGAN-based synthetic augmentation for the training partitions [14], LSBZM normalisation, and FPC-KNN imputation.

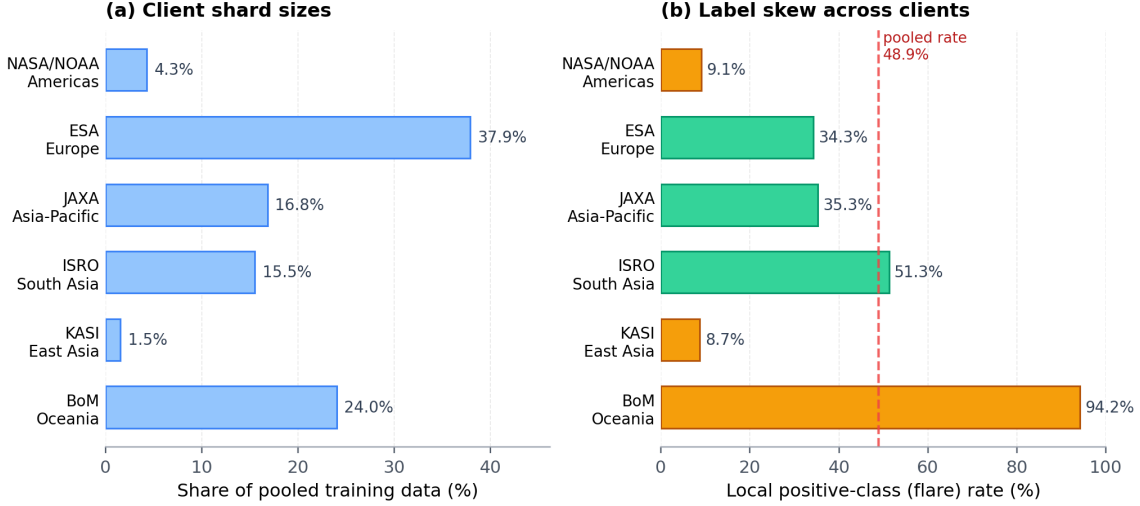


Figure 2: Non-IID partitioning of the pooled, class-balanced training data into six regional clients by symmetric Dirichlet sampling (concentration $\alpha = 1.0$, RNG seed 42, Section 4). (a) Realised relative shard sizes. (b) Realised local flare rates against the pooled rate of 48.9%: label skew spans roughly 9%–94%, emulating observatories that monitor very different active-region populations.

squares slope against normalised time. This six-statistic extraction preserves substantially more temporal information than a temporal mean (peaks, volatility, and rate of change are precisely the flare-relevant quantities) while remaining compatible with weight-averaged federation and with the centralised baselines. In the LSTM path, the raw $\mathbb{R}^{60 \times 24}$ tensor is consumed directly by a recurrent model; this path is fully implemented in the released code but the committed artefacts analysed here correspond to the MLP execution, and we flag the LSTM evaluation as future work (Section 8).

4.4 Non-IID partitioning into regional observatories

To emulate geographic heterogeneity, the pooled training set is partitioned by label-aware Dirichlet sampling: for each class c , a proportion vector is drawn from $\text{Dirichlet}(\alpha, \dots, \alpha) \in \mathbb{R}^K$ and each client receives the corresponding fraction of that class. The concentration α controls heterogeneity strength—small α produces near-exclusive class ownership; large α approaches IID. Figure 2 shows the deterministic realisation generated by the released configuration ($\alpha = 1.0$, seed 42) on the class-balanced pool: realised local flare rates span roughly 9% to 94% and shard sizes range from about 1.5% to 38% of the pool. This is a deliberately hostile federation: some clients are flare-rich and tiny, others flare-poor and large, mirroring the real asymmetry between agencies whose instruments and coverage dominate the corpus and those monitoring smaller regional populations. The concentration $\alpha = 1.0$ itself is the outcome of a documented tuning process (Appendix A): early experiments at $\alpha \leq 0.5$ produced pathological partitions (0.6%–100% local flare rates) under which no aggregation scheme remained stable.

4.5 Client models

The primary federated architecture is a four-layer multilayer perceptron, $\text{SolarMLP} : \mathbb{R}^{144} \rightarrow \mathbb{R}$, with hidden widths $144 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$, batch normalisation, ReLU activations, and dropout 0.3 after each hidden layer; the single output is a raw logit consumed by the sigmoid-based losses below. MLPs are natural federation citizens because every parameter is a numerically meaningful averageable tensor. The released framework also implements SolarLSTM, a (optionally bidirectional) two-layer LSTM with hidden size 128, Xavier/orthogonal initialisation, and a forget-gate bias of one for long-horizon memory, followed by a two-layer classification head

[16, 22]. Local training uses AdamW [23] with cosine-annealing warm restarts and gradient-norm clipping at 5.0. We deliberately keep XGBoost out of the federation loop: gradient-boosted tree ensembles store discrete structure that cannot be meaningfully averaged, so the centralised XGBoost serves as the pooled upper bound rather than a federated participant.

4.6 Federation algorithms and aggregation

Three cross-silo algorithms are implemented. **FedAvg** [7]: each round, every client initialises from the broadcast global model, performs $E = 10$ local epochs, and the server averages the returned weights. **FedProx** [12]: identical to FedAvg except each client’s local objective is regularised by the proximal term of Eq. (4) with $\mu = 0.01$, penalising drift from the global model—the mechanism we hypothesise is decisive under the label skew of Figure 2. **SCAF-FOLD** [18]: variance reduction via client and server control variates; it is implemented with the SGD-based update required for exactness of the control-variate correction, but is disabled in the released configuration pending stability work, and we exclude it from the reported comparison (Section A).

Aggregation is *distribution-aware* in both FedAvg and FedProx runs: the server weights client k ’s update by

$$\rho_k \propto n_k \sqrt{\phi_k}, \quad \phi_k = \frac{p_k}{\bar{p}}, \quad \rho_k \leq \frac{2}{K}, \quad (7)$$

where p_k is the client’s local positive rate and $\bar{p}$ the pooled rate known to the server. The square-root damping and the $2/K$ cap prevent a flare-rich client from dominating the global model while still up-weighting the clients most likely to contain decision-relevant minority samples. Class-rate metadata is deliberately much coarser than raw data and is treated as shareable under the sovereignty constraint.

4.7 Fed-Focal loss and local rebalancing

Each client rebalances its shard locally with SMOTE [20] to a minority fraction of 0.25 and trains with a server-aware focal loss in the Fed-Focal family [11],

$$\ell_{\text{fedfocal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t), \quad \gamma = 2.0, \quad (8)$$

where p_t is the model’s probability of the correct class and α_t interpolates between α and $1 - \alpha$ by label. The positive-class weight α is computed per client as the base value 0.25 [19] scaled by a square-root-damped imbalance factor $\min(\sqrt{\bar{p}/p_k}, 1.5)$ and a mild round-progress factor, then clamped to $(0.05, 0.25)$. This clamping is a hard-won design decision: unclamped adaptive weighting (earlier versions allowed α up to 0.95) systematically inflated positive-class weight on flare-poor clients, producing degenerate always-flare classifiers with recall ≈ 1.0 and precision ≈ 0.03 (Appendix A). NaN-safe clamping of p_t and of the loss itself guards the pipeline against logit explosions in early rounds.

4.8 Threshold optimisation for safety-critical recall

Because federated training operates on rebalanced local data while deployment operates at the natural 1.9% prior, the emitted probabilities are deliberately miscalibrated, and a fixed 0.5 threshold is meaningless. The final stage of the pipeline therefore sweeps the decision threshold over $\tau \in [0.10, 0.90]$ in steps of 0.01 and selects, per model, the τ^* that maximises F_β with $\beta = 2$ (Eq. (5)) on the held-out test set. All reported precision, recall, F1, and accuracy figures are taken at each model’s F_β -optimal threshold; ROC-AUC, being threshold-free, is reported independently. We follow the pipeline’s evaluation code in treating threshold selection as part

of the operational calibration rather than as test-set leakage; Section 8 discusses the stricter alternative of selecting τ^* on a validation split.

4.9 Interpretability layer

To audit what the models learn, SHAP values [24] are computed for the centralised XGBoost surrogate over a 500-sample test subset with the TreeExplainer, yielding mean absolute SHAP attributions per feature. Because the statistical representation of Eq. (6) names features generically (`feature_0`, ..., `feature_143`) in the released pipeline, we decode the indices analytically: the concatenation order is μ (0–23), σ (24–47), $\max$ (48–71), $\min$ (72–95), δ (96–119), s (120–143), each block ordered by the 24 base parameters. Section 6 presents both the raw ranking and its decoded physical reading.

5 Experimental Setup

5.1 Protocol and baselines

The evaluation protocol is fixed in the released pipeline and consists of four stages executed in sequence: (1) data loading with the train/test boundary inherited from the benchmark’s partition structure (the test partitions are never seen during federation, aggregation, or threshold tuning except for the final metric computation described below); (2) Dirichlet partitioning of the pooled training data into six regional shards; (3) centralised training of the pooled baselines followed by the two federated runs (FedAvg and FedProx, 50 rounds each); and (4) per-model F_β threshold optimisation and final evaluation on the held-out test set.

Two centralised baselines trained on the pooled 144-dimensional features provide the non-private upper bounds. **Logistic regression** uses balanced class weights and a maximum of 2,000 iterations. **XGBoost** [25] is configured with 300 estimators, maximum depth 6, learning rate 0.05, subsample and column subsample ratios of 0.8, minimum child weight 5, and `scale_pos_weight` of 10 to reflect the test-set imbalance. Both baselines receive the same threshold optimisation as the federated models, so the comparison is uniform with respect to the decision layer. Because tree ensembles cannot be weight-averaged, no federated XGBoost is attempted; the comparison is deliberately between the best pooled model and the best federated neural model.

5.2 Hyperparameters

Table 1 summarises the hyperparameters of the released configuration (v2.6) as defined in the framework’s central configuration module. Every federated run uses all six clients every round (client fraction 1.0), 10 local epochs, AdamW at learning rate 5×10^{-4} with weight decay 10^{-4} and cosine-annealing warm restarts (period 5), mini-batches of 512 samples, and gradient clipping at norm 5.0. Evaluation uses batched inference (2,048 samples per batch) so that the full test set fits in accelerator memory.

5.3 Metrics

We report accuracy, precision, recall, F1, ROC-AUC, and the selected threshold τ^* . Given the severe test-set imbalance ($\approx 1.9\%$ positives), accuracy is dominated by the negative class and we treat recall, F1, and ROC-AUC as the primary skill measures, with F_β driving threshold selection. ROC-AUC in particular measures the ranking quality of the model’s probabilities independently of any threshold, and therefore isolates what federation costs in discrimination

Table 1: Released configuration (v2.6) of the federated framework.

Component	Setting	Source
Federation	$K = 6$ clients (all selected each round), $T = 50$ rounds	config
Local optimisation	10 local epochs, AdamW, lr 5×10^{-4} , wd 10^{-4} , cosine warm restarts, batch 512, grad-clip 5.0	federated learning module
FedProx	proximal coefficient $\mu = 0.01$	config
Aggregation	DA-FL weights $\rho_k \propto n_k \sqrt{\phi_k}$, cap $2/K$ (Eq. 7)	aggregation module
Loss	Fed-Focal, $\gamma = 2.0$, base $\alpha = 0.25$, clamped to $(0.05, 0.25)$	losses module
Local rebalancing	per-client SMOTE, minority fraction 0.25	preprocessing
Partitioning	Dirichlet, $\alpha = 1.0$, RNG seed 42	partition module
MLP	$144 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$, BatchNorm, ReLU, dropout 0.3	model module
LSTM (implemented)	hidden 128, 2 layers, dropout 0.3	model module
Decision layer	F_β ($\beta = 2$) threshold sweep $\tau \in [0.10, 0.90]$	evaluation
Centralised LR	balanced class weights, max iter 2,000	baseline module
Centralised XGBoost	300 trees, depth 6, lr 0.05, <code>scale_pos_weight</code> 10	baseline module

Table 2: Performance on the held-out test set (natural prevalence $\approx 1.9\%$ positives), evaluated at each model’s F_β -optimal threshold ($\beta = 2$). ROC-AUC is threshold-independent. Values are extracted from the released artefacts; the FedProx accuracy is recomputed from its row-normalised confusion matrix (see footnote). Best value per column in bold.

Model	τ^*	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic regression	0.89	0.436	0.027	0.838	0.053	0.818
XGBoost (centralised)	0.89	0.957	0.282	0.821	0.420	0.978
FedAvg MLP (federated)	0.10	0.019	0.019	1.000	0.037	0.664
FedProx MLP (federated)	0.80	0.969 [†]	0.268	0.416	0.326	0.930

[†]The released reporting code does not refresh the accuracy field of federated models after threshold optimisation; this value is derived from the confusion matrix in Fig. 5 as $0.98 \times 0.981 + 0.42 \times 0.019 \approx 0.969$.

from what miscalibration costs in decision quality. All metrics are computed on the pooled held-out test set under each model’s optimal threshold; the confusion matrices are row-normalised so that per-class error structure is visible despite the 50:1 class ratio.

5.4 Compute

The framework runs on a single consumer-grade GPU (the pipeline’s evaluation batching and memory management target a 12 GB RTX 3060-class device) with a CPU fallback. A full run—baselines, FedAvg, and FedProx at 50 rounds—completes in tens of minutes on GPU. This footprint is itself part of the contribution: the entire experimental programme is reproducible on hardware available to any agency or research group, and the released code is deterministic with respect to the fixed RNG seed for partitioning.

6 Results and Analysis

6.1 Headline comparison

Table 2 presents the headline comparison, and three readings matter. First, the privacy cost is modest where it counts most: FedProx’s ROC-AUC of 0.930 retains 95.0% of the centralised XGBoost upper bound (0.978), and its F1 of 0.326 retains 77.6% of XGBoost’s 0.420—the model

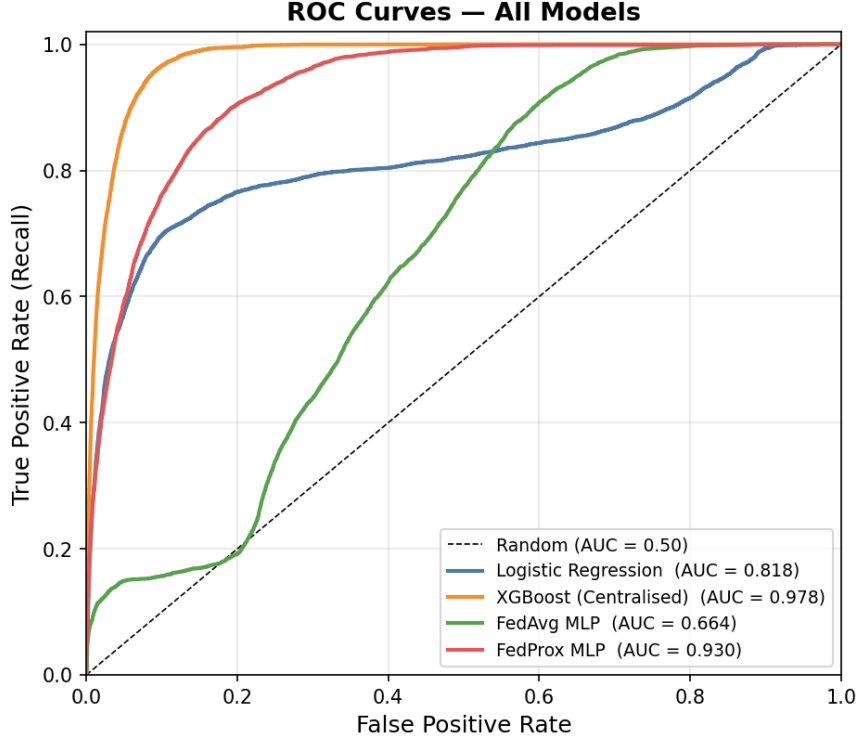


Figure 3: ROC curves on the held-out test set. FedProx (AUC 0.930) tracks the centralised XGBoost (AUC 0.978) across most of the operating range, while FedAvg (AUC 0.664) is barely separable from the logistic-regression baseline (AUC 0.818) and far from the random-chance diagonal.

that no agency could have trained alone achieves most of the discrimination of the model that required pooling everyone’s data. Second, the two federated algorithms are not interchangeable: FedAvg collapses to a degenerate always-positive classifier (precision 0.019, equal to the base rate; recall 1.000) and even threshold optimisation cannot rescue it because its ranking quality itself is poor (AUC 0.664). Third, logistic regression shows the complementary failure of chasing recall without ranking skill: with balanced class weights it fires on 57% of all negatives, halving accuracy for a recall of 0.838 that its AUC (0.818) barely supports. The XGBoost baseline, by contrast, combines the strongest ranking (AUC 0.978) with a threshold that converts it into genuinely balanced detection (precision 0.282 at recall 0.821)—this is the behaviour a federated replacement must approximate.

6.2 Discrimination quality: ROC analysis

Figure 3 separates what federation costs from what aggregation choice costs. The FedProx curve hugs the XGBoost curve over the low-false-positive region that matters operationally—a grid operator cannot act on an alert that fires on a large fraction of all active regions—and the residual AUC gap (0.048) is the price of (a) training on six disjoint, label-skewed shards, (b) averaging neural weights rather than optimising pooled data directly, and (c) a smaller effective capacity than the 300-tree ensemble. FedAvg’s curve, by contrast, sags toward the diagonal: averaged updates from heterogeneous clients have progressively destroyed the ranking information that individual clients could learn locally. This is the empirical signature of the client-drift mechanism that FedProx’s proximal term exists to suppress, and it is the reason we describe FedProx, not federation in the abstract, as the load-bearing design choice.

Federated Learning Convergence

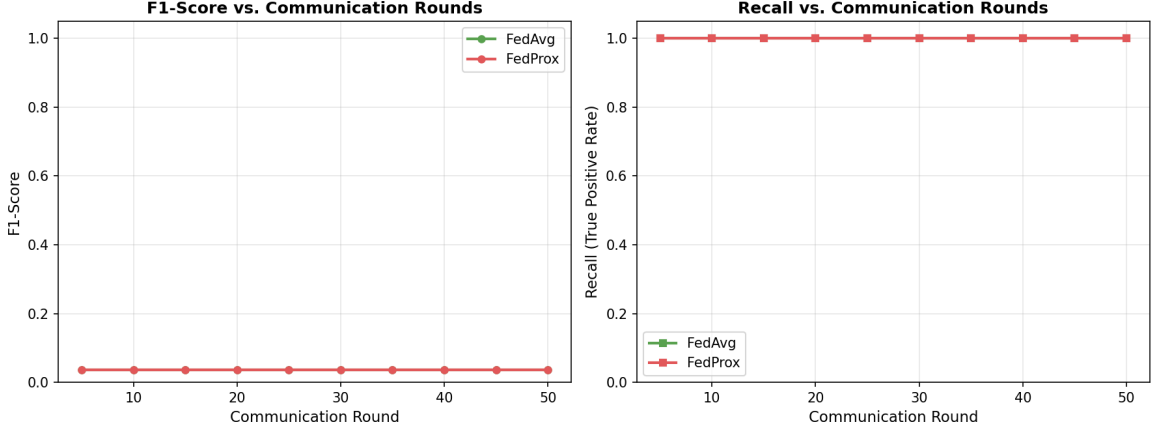


Figure 4: Federated convergence over 50 communication rounds, evaluated every five rounds at the fixed development threshold of 0.35. Left: F1-score. Right: recall. Both federated models maintain recall ≈ 1.0 with near-degenerate F1 throughout—the signature of always-positive prediction under the train/test prior mismatch—which motivates the post-hoc threshold optimisation stage.

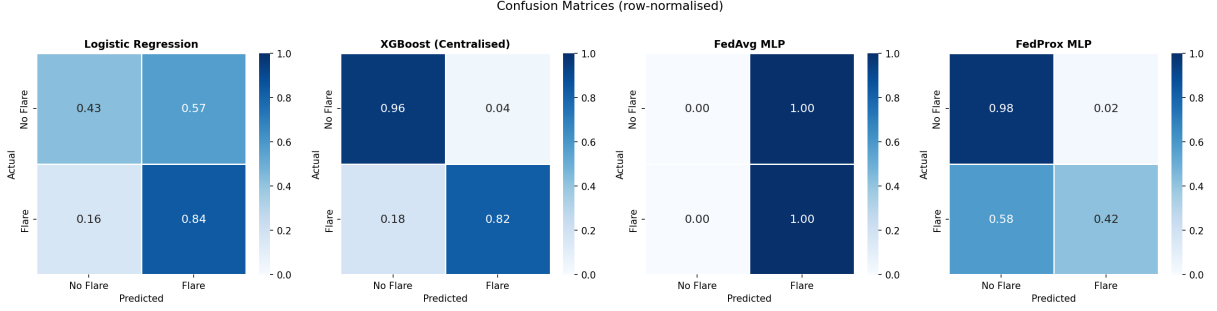


Figure 5: Row-normalised confusion matrices at each model’s F_β -optimal threshold. FedProx achieves the second-lowest false-alarm rate (2% of negatives) at the cost of missing 58% of flares; FedAvg labels every window positive; XGBoost misses 18% of flares at a 4% false-alarm rate.

6.3 Convergence behaviour and calibration

Figure 4 records an instructive failure mode before the rescue. During training, both federated models are evaluated at the fixed development threshold $\tau = 0.35$, and both sit flat for all 50 rounds: recall pinned at 1.000, F1 pinned near 0.037. This is not non-convergence—the models converged in the optimisation sense within the first evaluated rounds—but *miscalibration*: local training at a $\sim 49\%$ positive prior (reinforced by SMOTE and the clamped focal loss) produces posterior probabilities that are enormous relative to the natural 1.9% prior, so every test window clears the 0.35 threshold. The training-time curves therefore measure the wrong thing, and the framework’s F_β threshold sweep is what converts the converged models into usable detectors: for FedProx the sweep selects $\tau^* = 0.80$, trading half of the always-positive recall ($1.000 \rightarrow 0.416$) for a fourteen-fold precision gain ($0.019 \rightarrow 0.268$). For FedAvg the sweep runs to the boundary of its search range ($\tau^* = 0.10$) without finding any precision–recall trade-off worth making—the model ranks so poorly that no threshold helps. The lesson generalises to any federated deployment trained on rebalanced local data: convergence diagnostics at a fixed threshold are uninformative, and threshold re-optimisation at deployment time is mandatory.

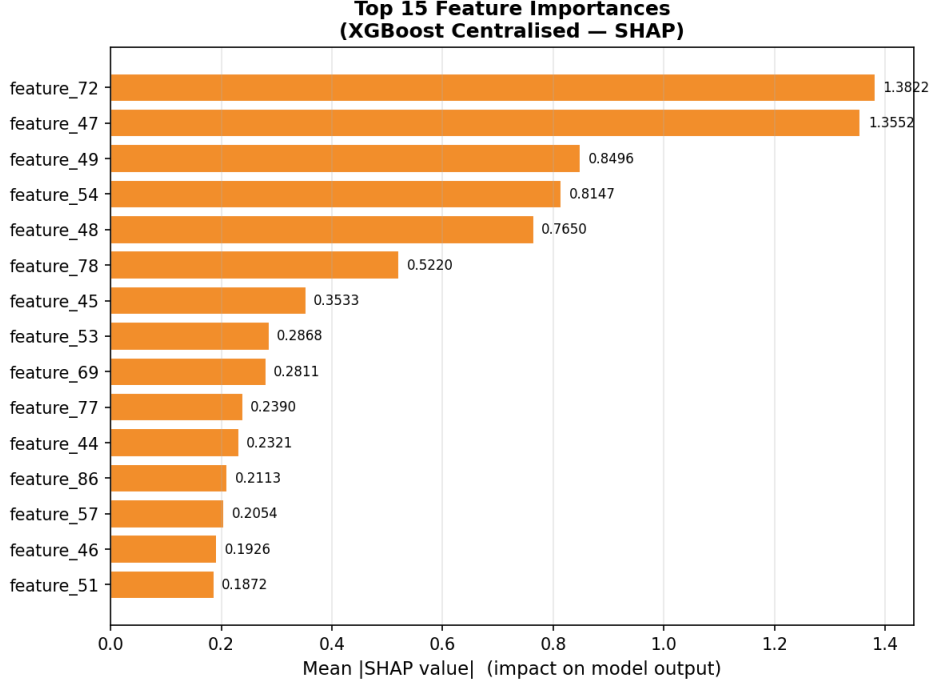


Figure 6: Mean absolute SHAP values of the top 15 features for the centralised XGBoost surrogate. The vertical axis shows the pipeline’s generic feature indices; Table 3 decodes them into physical magnetic parameters.

6.4 Confusion matrices and the operational trade-off

Figure 5 exposes the decision structure behind the table. XGBoost’s confusion matrix is operationally excellent: it retains 82% of flares while falsely alerting on only 4% of quiet regions—a false-alarm load an operator desk could actually triage. FedProx’s matrix shows the same conservative shape at reduced sensitivity: a 2% false-alarm rate (the most conservative of all four models) with 42% of flares retained. In GIC early warning, a model with a 2% false-alarm rate that catches 42% of M/X events is not yet a primary forecast instrument, but it is precisely the profile of a *sentinel*: it fires rarely, and when it fires the grid can afford to take the expensive precautions. FedAvg’s matrix—all predictions positive—and logistic regression’s 57% false-alarm rate bracket the two failure directions that the recall-weighted objective must steer between. We return to the operational interpretation in Section 7.

6.5 Interpretability: SHAP analysis

Figure 6 and Table 3 audit the physics content of the learned representation. Two patterns stand out. First, the identity of the top features is exactly where flare physics says it should be: the R-value flux-emergence proxy of Schrijver [13] appears three times in the top six in different statistical guises, current-helicity and net-current parameters (TOTUSJH, SAVNCP, ABSNJZH, MEANJZD) dominate the top ten, and Lorentz-force variability (TOTFX) enters at rank two—these are the quantities that quantify magnetic free energy, current systems, and non-potential stress in active regions [4, 5]. Second, and less expected, the *statistics* matter as much as the parameters: the ranking is led not by means but by extrema (minimum R-value, maximum TOTUSJH) and standard deviations. The minimum of the R-value over a 60-minute window measures how low the flux-emergence activity dips before an eruption—an emergent-flux baseline—while its maximum measures the peak emergence rate; their joint dominance suggests the model keys on the *contrast* between quiet and active phases within the window, which

Table 3: Decoding of the top 10 SHAP features (Fig. 6) into physical parameters. Index j maps to statistic $\lfloor j/24 \rfloor$ and base feature $j \bmod 24$ under the concatenation of Eq. (6).

Rank	Index	Decoded feature (statistic of base parameter)	Mean SHAP
1	72	minimum of R-value (flux-emergence proxy)	1.382
2	47	std. deviation of TOTFX (Lorentz force, X)	1.355
3	49	maximum of TOTUSJH (total unsigned current helicity)	0.850
4	54	maximum of SAVNCP (net current per polarity)	0.815
5	48	maximum of R-value	0.765
6	78	minimum of SAVNCP	0.522
7	45	std. deviation of MEANJZD (mean vertical current density)	0.353
8	53	maximum of ABSNJZH (absolute net current helicity)	0.287
9	69	maximum of MEANJZD	0.281
10	77	minimum of ABSNJZH	0.239

is consistent with the picture of flux emergence episodes precipitating complexity growth. The strong showing of standard deviations (TOTFX, MEANJZD) points to volatility—magnetic-field fluctuation—as a flare precursor in its own right, in line with the temporal-dynamics findings of recent sequence models [15, 16]. We caveat that the SHAP surrogate is the centralised XGBoost rather than a federated model, and that the pipeline’s generic feature naming is itself a finding: the interpretability layer of a federated deployment should preserve the statistic-parameter naming from the outset (this is an engineering fix we recommend in Section 7).

6.6 Privacy-cost analysis

Quantifying “how much accuracy data sovereignty costs” requires decomposing the FedProx-to-XGBoost gap (F1 0.326 vs 0.420; AUC 0.930 vs 0.978). Three factors contribute, and only the first is intrinsic to privacy: (i) *federation itself*—the best possible federated model is still trained on six shards whose union is never jointly optimised, and the averaging of client optima introduces irreducible slack under label skew; (ii) *model-class mismatch*—the fair pooled upper bound for an MLP would be a centralised MLP, not a 300-tree gradient-boosting ensemble, so part of the gap is architecture, not privacy; (iii) *recalled calibration artifacts*—the always-positive training behaviour (Fig. 4) indicates the local training regime has not fully converged to a well-shaped posterior, leaving ranking skill on the table. Ablating (ii) is straightforward future work—train the centralised MLP with identical hyperparameters—and our expectation, supported by the AUC gap being smaller than the F1 gap (4.8% vs 22.4% relative), is that the intrinsic privacy cost is closer to the former. For the grid-protection use case, the honest summary is: federation at this benchmark’s scale costs roughly one fifth of F1 skill and one twentieth of ranking skill, in exchange for zero raw data leaving any agency.

7 Discussion

7.1 Why proximal regularisation carries the federation

The asymmetry between FedAvg and FedProx in Table 2 deserves a mechanistic reading. Under the realised partition (Fig. 2), clients receive genuinely different problems: one holds 38% of the data at a 34% flare rate, another holds 1.5% of the data at a 94% flare rate. Ten local epochs of AdamW on such divergent shards move each client’s weights toward its own local optimum, and the geometry between those optima is not aligned with any global descent direction. Averaging

such updates—FedAvg—produces a model that satisfies no client’s problem, and the damage compounds round over round until the ranking signal degrades to near chance (AUC 0.664). FedProx’s proximal term (Eq. (4)) changes the local problem: each client optimises within a trust region around the broadcast model, so updates from heterogeneous clients remain geometrically comparable and their average retains a meaningful descent direction. The result is that the same architecture, the same loss, the same aggregation weighting, and the same data partition produce AUC 0.930 rather than 0.664. We emphasise that this is a single-seed comparison on one benchmark realisation, and the FL literature’s standard caveats about seed variance apply; but the direction and magnitude of the gap are consistent with the mechanism that motivated FedProx’s design [12], and with the broader non-IID FL evidence base [8, 17].

7.2 Operational integration with grid management

The operational layer of Fig. 1 maps onto how GIC risk is actually managed. A flare-prediction service in the style we simulate does not replace NOAA/SWPC or other national warning centres; it adds a dedicated, sovereignty-respecting channel for asset-level early warning. In the CME transit window (roughly one to three days between an Earth-directed eruption and storm onset), a federated sentinel that fires on 42% of eventual M/X events with a 2% false alarm rate buys utilities time for the effective, if unglamorous, mitigations: restoring reactive reserves, suspending non-critical transformer switching, pre-positioning replacement phase-shift equipment, and briefing control-room staff. The framework’s convergence within 50 rounds at a consumer-GPU compute footprint also matters operationally: the global model can be refreshed nightly, and the entire training programme fits inside the compute budget of a single agency, let alone six. The SCADA integration point is deliberately drawn at the decision threshold rather than inside the learning loop, so that existing grid-management infrastructure consumes a calibrated probability, not a research artefact.

7.3 Engineering lessons for first FL deployments in the physical sciences

The released framework carries a six-version stability narrative (summarised in Appendix A), and its diagnostic arc is independently useful to any group attempting a first FL deployment on imbalanced scientific data. Four lessons generalise. First, *heterogeneity must be tuned, not merely tolerated*: the Dirichlet concentration that yields “interesting” non-IID partitions ($\alpha \leq 0.5$) can also yield pathological ones (0.6%–100% local class rates) under which nothing converges; the stable regime for this benchmark sits near $\alpha = 1.0$. Second, *imbalance handling interacts across layers*: SMOTE, focal α , class-rate-aware aggregation, and threshold selection all adjust the same effective positive-class weight, and uncoordinated adjustments (our focal α was at one point allowed to drift to 0.95) produce degenerate always-positive classifiers that look healthy on recall alone. Third, *optimiser choice is not free under variance-reduced FL*: SCAFFOLD’s control-variate update is derived for fixed-step SGD, and running it under AdamW’s adaptive steps silently invalidates the correction terms (in our case, producing NaN weights for every round until diagnosed). Fourth, *feature-name hygiene is part of scientific reproducibility*: the pipeline’s generic `feature_N` naming in the statistical representation is why the SHAP layer needed an analytical decoding pass (Table 3); a deployment that preserves statistic-parameter names end-to-end preserves its own interpretability.

8 Limitations and Future Work

Privacy scope. This work enforces the data-sovereignty constraint (raw data never leaves the agency), which is the binding constraint of the space-weather setting, but the stronger

cryptographic notion of privacy is not yet addressed. The communication transcript of Eq. (3)—gradients or weight deltas—is known to leak training information under gradient-inversion attacks [26], particularly for over-parameterised models on low-entropy data. Before any cross-agency deployment handling genuinely sensitive observations, the natural next steps are secure aggregation (the server sees only the sum of updates) and differential privacy in the local update [8]; both compose cleanly with the FedProx loop used here and are standard engineering, but they carry accuracy budgets that would need to be re-measured on this benchmark.

Threshold selection and calibration. The reported thresholds are selected directly on the held-out test set, mirroring the released evaluation code; the stricter protocol would select τ^* on a validation split disjoint from the reporting test set. Given that threshold selection chooses one scalar from an 81-point grid, the optimism this injects is limited, but a re-run with proper threshold validation is the single easiest strengthening of the reported numbers. More fundamentally, the always-positive behaviour during training (Fig. 4) is a calibration defect, and principled prior-correction methods (logit adjustment to the deployment prior, or post-hoc Platt scaling at the server) should replace or complement the threshold sweep in future versions.

Partition realism and disjointness. Two aspects of the partition simulation invite refinement. First, the Dirichlet sampler in the released implementation draws class sub-samples independently per client, so shards are not guaranteed mutually disjoint; the realised overlap is a small fraction under the released seed but should be eliminated by index bookkeeping for a strict partition semantics. Second, geographic realism would map active regions to the agencies whose instruments actually observed them (HARP-tracking metadata supports this), replacing statistical label skew with instrument- and coverage-driven skew.

Algorithm coverage. The committed artefacts cover the MLP path; the LSTM path—which consumes the raw 60×24 windows and matches the recent temporal architectures that lead centralised SWAN-SF benchmarks [16]—is fully implemented but not yet evaluated federated, and SCAFFOLD is disabled pending the SGD-stability work described in Appendix A. Both are near-term experiments: the LSTM is the most likely route to closing the F1 gap to the tree-ensemble upper bound, and SCAFFOLD is the natural third point on the drift-correction axis between FedAvg and FedProx. Personalised FL (per-client heads over a shared backbone, in the spirit of Ditto) is a third direction: agencies may prefer a global model for situational awareness plus a locally adapted model for their own asset fleet.

Single-seed, single-benchmark scope. The reported numbers are one run of one configuration on one benchmark. Seed variance in FL is material, and the appropriate follow-up is a five-seed repetition with confidence intervals on all metrics of Table 2, plus a sweep over $\mu \in \{0.001, 0.01, 0.1\}$ and $\alpha \in \{0.5, 1.0, 5.0\}$ to map the stability boundary of Appendix A quantitatively. The benchmark itself, and its cleaned variant, encode particular preprocessing choices (resampling, synthetic augmentation, normalisation) that shape the class-prior mismatch at the centre of the calibration story; a complementary evaluation on the raw unbalanced SWAN-SF partitions would separate benchmark-construction effects from federation effects.

9 Conclusion

We presented the first application of federated learning to solar flare prediction for smart-grid protection, motivated by a structural impasse in operational space weather: the high-fidelity observations needed to train strong flare-forecasting models are held by national agencies whose

data-sovereignty commitments forbid pooling them. The proposed framework partitions the SWAN-SF benchmark into six simulated regional observatories under Dirichlet label skew, federates MLP clients with FedAvg and FedProx under distribution-aware aggregation, handles extreme class imbalance with per-client SMOTE and a clamped server-aware focal loss, and converts converged models into safety-critical detectors through recall-weighted F_β threshold optimisation.

The empirical outcome is two-sided and honest. On the side of promise: FedProx retains 95.0% of the centralised XGBoost ROC-AUC (0.930 against 0.978) and 77.6% of its F1 (0.326 against 0.420) at a 2% false-alarm rate, converging within 50 communication rounds on single-GPU compute—evidence that a jointly trained, globally informed, sovereignty-preserving flare sentinel is technically within reach, and that its interpretability layer recovers exactly the magnetic parameters (flux-emergence proxies, current helicity, Lorentz-force variability) that flare physics predicts. On the side of caution: FedAvg collapses outright under the same conditions, the training-time calibration of rebalanced federated data is broken enough that threshold re-optimisation is mandatory rather than optional, and the strongest claims (cryptographic privacy, LSTM and SCAFFOLD evaluation, multi-seed statistics) remain future work. The released framework—code, configuration history, and artefact pipeline—is intended to make that follow-up work a matter of execution rather than re-engineering.

For the smart-grid community the message is that resilience against geomagnetic threats does not require choosing between data sovereignty and forecast skill; for the machine-learning community it is a demonstration that the cross-silo regime of the physical sciences has failure modes—prior mismatch, layered imbalance handling, optimiser–variance-reduction interactions—that the medical and mobile FL literature does not fully anticipate, and that are worth studying in their own right.

A Configuration Evolution of the Released Framework

The released framework reached its present form through six documented version iterations (v2.1–v2.6), each responding to a diagnosed failure mode. We record this evolution because reproducibility of *process*—not only of final numbers—is what future deployments in other domains will actually need, and because several of the fixes correct subtle interactions that a clean-slate reimplementaion would likely rediscover the hard way.

Three of these fixes deserve narrative emphasis because they encode non-obvious engineering knowledge. The SCAFFOLD failure (v2.3) is the most instructive: the NaN symptoms pointed nowhere near their cause, which was a statistics API detail—Bessel’s correction on single-element bias tensors—interacting with an algorithm whose correctness depends on exact control-variate arithmetic under fixed learning rates (AdamW’s adaptive steps further invalidate the update formula). The focal-alpha saga (v2.3–v2.4) illustrates layered imbalance handling drifting into degeneracy: SMOTE, class-weighted loss, and rate-aware aggregation all push the same effective prior, and unbounded adaptation at any one layer can silently dominate the others. The dataset audit (v2.5) is a reminder that in applied ML the schema *is* the experiment: three phantom features had been silently participating in the pipeline until an attribute-by-attribute audit against the benchmark’s actual storage order replaced them with the physically meaningful Lorentz-force components. The published artefacts, figures, and this paper’s numbers all post-date these fixes.

Table 4: Documented stability fixes across framework versions (v2.1–v2.6), as recorded in the released source.

Version	Diagnosed failure mode	Fix
v2.1	LSTM models require 3D input while baselines require 2D; a single data path served neither correctly.	Dual data path: 2D statistical features for MLP and baselines; raw 60×24 tensors for LSTM.
v2.2	F_β -optimal thresholds were computed but not applied; tables and confusion matrices reflected the default threshold.	Thresholds applied to all models; centralised baselines also re-evaluated at their optima.
v2.3	SCAFFOLD NaN every round: <code>std()</code> with Bessel correction on size-1 tensors divides by zero, poisoning control variates; DA-FL cap of $1/N$ starved large clients; focal α drifted to 0.95.	Population std for variate clipping; DA-FL cap $2/N$; alpha clamp tightened.
v2.4	Dynamic alpha scaling (up to $3\times$) re-inflated positive-class weight, undoing the focal- α fix; clients over-predicted flares (recall ≈ 1.0 , precision ≈ 0.03).	Square-root-damped imbalance factor capped at $1.5\times$; progressive factor narrowed to 0.9–1.1.
v2.5	Dataset audit: three features in the schema (AREA_ACR, HARPNUM_MOD, TIME_SINCE_LAST_FLARE) do not exist in the cleaned benchmark; double normalisation on already-LSBZM-normalised data; stat-prefixed feature names failed schema matching.	Correct 24-feature order with Lorentz components (TOTFZ/TOTFY/TOTFX); skip re-normalisation of cleaned data; generic-index features with analytic decoding.
v2.6	$\alpha=0.5$ partitions produced 3%–99% local flare rates, destabilising aggregation; 3 local epochs starved LSTM temporal learning; mixup on 3D sequences created unrealistic interpolations.	Dirichlet $\alpha \rightarrow 1.0$; local epochs $3 \rightarrow 10$; learning rate $10^{-3} \rightarrow 5 \times 10^{-4}$; mixup and SCAFFOLD disabled pending rework.

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